

Richard Goodman / apoth3osis — Meet-and-Greet Agenda

Date: May 2026 **Primary Contact:** Richard Goodman, Apoth3osis Labs — rgoodman@apoth3osis.io

Website: <https://www.apoth3osis.io> — “Paper → Proof → Code”

1. Objective

Explore formal verification of core QWAV claims in Lean 4:

1. **The strong triangle inequality produces passive error confinement** in Bruhat-Tits tree encodings
2. **A threshold theorem for ultrametric quantum error correction** — analogous to the surface code threshold theorem, but geometric rather than measurement-based
3. **Energy barrier scaling** — exponential growth with tree depth

A Lean proof that type-checks is unassailable — no reviewer, no gatekeeper.

2. Collaboration Shapes

Based on your email proposal, three shapes on the table:

#	Shape	Output
1	Formalization transport	Co-authored paper + Lean repo proving specific QWAV claims
2	Ultrametric triangulation	Three-way program: QWAV p-adic + Veselov subspace lattices + HeytingLean topos
3	Position paper	Substrate computation: LoF-to-physics bridge

Shape #1 seems like the natural starting point — concrete, scoped, produces a verifiable artifact.

3. What QWAV Brings

Asset	Status
Bruhat-Tits tree data structure	Implemented in Python
Strong triangle inequality	0 violations in 15,000 trials
Error confinement	Tree LER=0 at depth 3+ for physical error rates up to 40%
Energy barrier scaling	Barrier = 2^{depth} , verified exhaustively for small depths
Published experimental parameters	Google/IBM/Quantinuum qubit coherence times, gate fidelities
UQC Release document	57 citations, full mathematical definitions

4. Core Theorem to Formalize

Threshold Theorem for Ultrametric QEC (informal): In a Bruhat-Tits tree encoding of depth d over p -adic numbers, with physical error rate per leaf qubit, the logical error rate at the root is bounded by $C \times (\text{physical error rate})^{(k \times d)}$ for constants C and k determined by tree structure and error model.

Corollary: There exists a threshold error rate such that for all physical error rates below it, the logical error rate approaches zero as tree depth increases. This threshold is geometric, not measurement-based.

This is formally verifiable (trees and probability bounds are definable in Lean), parameterizable with published experimental data, and publishable regardless of whether the bound holds.

5. Proposed Deliverable

Title: *A Threshold Theorem for Ultrametric Quantum Error Correction: Formal Verification in Lean 4*

Content: 1. Formal definition of Bruhat-Tits tree encoding in Lean 4 2. Statement and proof of the strong triangle inequality for tree-encoded states 3. Derivation of error confinement bound 4. Proof of exponential suppression with tree depth 5. Parameterization with published experimental data 6. Machine-verified repository (zero “sorry”)

Output: Co-authored paper + Lean repository (arXiv/Zenodo).

Timeline: 2–3 months for first theorem. Each additional claim is a new module.

Roles: - **Rowan:** Provide mathematical claims, tree structure definitions, error model specifications, experimental parameter references - **Richard:** Translate claims into Lean 4, build verified repository - **Joint:** Write the paper, agree on scope

6. Agenda

Opening (5 min)

State the goal: explore Shape #1 formalization transport for the ultrametric QEC threshold theorem.

What We’ve Built (10 min)

- Tier 0 computational validation: 9 Python files demonstrating error confinement and barrier scaling
- Tree encoding: zero logical error at depth 3+ for physical error rates up to 40%, vs. flat encoding at 15% LER
- Energy barrier scales as 2^{depth}
- Strong triangle inequality: zero violations in 15,000 random trials

Core Theorem (10 min)

Present the threshold theorem concept. Discuss formalizability, parameterizability, and publishability.

The Ask (10 min)

1. Interest in formalizing this as Shape #1?
 2. Scope: start with strong triangle inequality $\rightarrow$ error confinement proof
 3. Timeline: Lean module would be "Generative/QuniAxiomatic.lean"
 4. What Richard needs from us: formal theorem statements, tree definitions, error models, experimental parameters
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7. What to Have Ready

- Tier 0 simulation results (can screen-share)
 - UQC Release document (57 citations)
 - Formal theorem statement
 - Published experimental parameters with citations
 - This agenda
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Shared ahead of time so we can spend the call on the actual conversation.